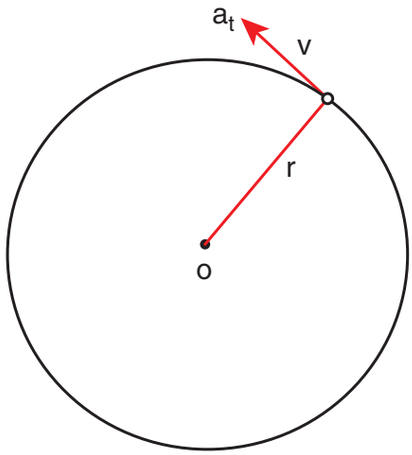


Figure 2.48 Tangential acceleration

Note that the tangential acceleration is in the direction of linear velocity.

2.11.6 Circular Motion

When a point object is moving on a circular path with a constant speed, it covers equal distances on the circumference of the circle in equal intervals of time. Then the object is said to be in uniform circular motion. This is shown in Figure 2.49.

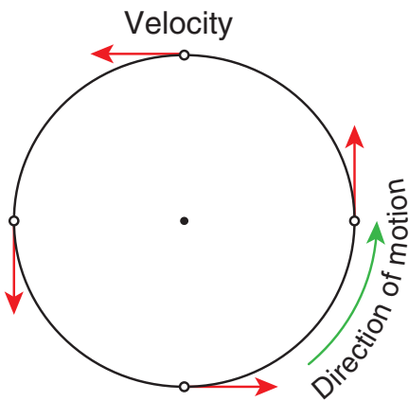


Figure 2.49 Uniform circular motion

In uniform circular motion, the velocity is always changing but speed remains the same. Physically it implies that magnitude of velocity vector remains constant and only the direction changes continuously.

If the velocity changes in both speed and direction during the circular motion, we get non uniform circular motion.

Centripetal acceleration

As seen already, in uniform circular motion the velocity vector turns continuously without changing its magnitude (speed), as shown in Figure 2.50.

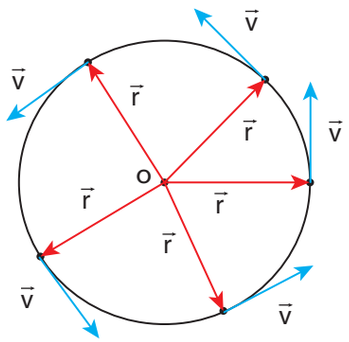


Figure 2.50 Velocity in uniform circular motion

Note that the length of the velocity vector (blue) is not changed during the motion, implying that the speed remains constant. Even though the velocity is tangential at every point in the circle, the acceleration is acting towards the centre of the circle. This is called centripetal acceleration. It always points towards the centre of the circle. This is shown in the Figure 2.51.

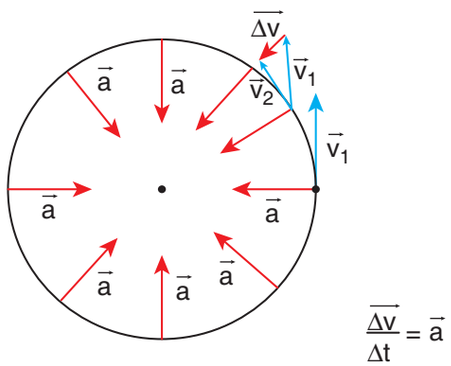


Figure 2.51 Centripetal acceleration

UNIT-2(XI-Physics_Vol-1)-1.indd 91 05-01-2022 22:15:49